PROXMOX HOMELAB · ENGINEERING REPORT

Homelab Infrastructure & Operations

Service modernization, private media access, dashboard visibility, and measured storage recovery on a self-hosted Proxmox platform.

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Docker containers running at inspection

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Service endpoints responding to HTTP checks

3
Storage cleanup operations with verified recovery

- Proxmox VE & Docker Compose
- Jellyfin & media services
- Tailscale & mobile access
- Homarr & operational visibility
- Migration & legacy retirement
- Storage & backup management

Open operational item

Local guest backups require additional destination capacity. The latest successful job in the inspected history ran on November 9.

Reporting date · November 11, 2026, Eastern time
Latest availability checks · 11:15 p.m. Eastern time

COMPUTE, SERVICES & ACCESS

One host. Defined service roles.

The media applications run in Docker inside an unprivileged Linux container. Wazuh occupies a separate virtual machine on the Proxmox host.

Phone or computer → Tailscale → private service access

Proxmox VE host

Media LXC · 4 cores · 8 GiB RAM

Homarr dashboard	Jellyfin playback	Seerr requests

Library management → download client → shared media filesystem

Sonarr / Radarr Prowlarr / Bazarr	qBittorrent Gluetun networking	Media volume 250 GiB allocated

Wazuh VM 4 cores · 8 GiB RAM · 40 GiB disk	Local guest backup destination Capacity-guarded scheduled jobs

SERVICE LAYER	DEPLOYED COMPONENTS
Playback & requests	Jellyfin, Seerr
Library management	Sonarr, Radarr, Prowlarr, Bazarr, Recyclarr
Downloads & networking	qBittorrent, Gluetun
Supporting services	Byparr, Cleanuparr
Dashboard & Docker visibility	Homarr, socket-proxy

The LXC has a 32 GiB system volume. TUN and render device assignments support private networking and media workloads. Performance validation requires an appropriate playback test.

ACCESS, MIGRATION & USABILITY

From native services to an integrated stack.

REMOTE ACCESS REPAIR

TUN passthrough restored Tailscale operation

The unprivileged media container lacked the networking device Tailscale needed. Device passthrough and a container stop/start enabled the daemon. Account authorization completed the connection.

Verification · Tailscale connectivity, Jellyfin health response, and HTTP 200 from the web interface.

DOCKER MIGRATION

Media applications moved into Compose

Jellyfin, Radarr, Sonarr, Prowlarr, and qBittorrent moved from native services into Docker. Dedicated application data and a shared media filesystem supported the new layout.

Recovery provision · The pre-migration Proxmox snapshot was retained.

DASHBOARD IMPROVEMENTS

Homarr became the mobile entry point

Quick links and download progress lead the phone layout. Playback, calendar, requests, storage, backup status, Proxmox resources, and Docker state provide operational context.

Verification · Eight integrations passed connection checks. Desktop and iPhone-sized browser checks passed.

Shared media filesystem

Downloads and library imports share a filesystem, supporting hardlinks where configured. Application data uses dedicated directories.

Private administration

Tailscale provides service access from authorized clients. qBittorrent shares Gluetun's network namespace.

Live operational cards

The storage card reads filesystem capacity through Radarr every minute. The backup card reads Proxmox job and archive information every five minutes.

APPLICATION RETIREMENT & MEDIA CLEANUP

Space recovered. Results verified.

CONTAINER SYSTEM FILESYSTEM

1.22 GiB

Retired native installations removed

Removed the original application binaries, obsolete service definitions, packages, and unused data directories after reviewing the live Docker mounts.

13 containers stayed running

MEDIA FILESYSTEM

19.32 GiB

Three movies removed across two batches

Used Radarr's file deletion and import exclusions, checked remaining records, and refreshed Jellyfin to reconcile the library catalog.

132.18 GiB free afterward

Verified removal batches

Initial cleanup · 15.16 GiB recovered. Follow-up removal · 4.17 GiB recovered. The combined figure uses the underlying byte measurements.

Identify	Matched the requested entries to their exact records and files. Each media file had one filesystem link.
Reconcile copies	Checked the active download list and filesystem. No matching download copies remained.
Remove & refresh	Waited for asynchronous file deletion to finish, then confirmed both entries were absent from Jellyfin.
Verify scope	The first batch preserved 20 other Radarr records. The follow-up preserved the remaining 19. All nine endpoint checks passed.

Storage accounting

These results measure recovery inside separate filesystems. Retained Proxmox snapshots can still reference removed blocks on the storage pool. Physical pool recovery requires its own measurement.

LOCAL GUEST BACKUP INCIDENT

A capacity guard stopped the backup.

The scheduled local guest job targets the media LXC and Wazuh VM. Its preflight guard estimates capacity from allocated guest blocks, adds 25% headroom, and reserves 10 GiB for the host.

NOVEMBER 10 INCIDENT FIGURES

63.0 GiB available **168.5 GiB required**



Available destination capacity covered approximately 37% of the guard's estimated requirement.

SCHEDULED RUN · EASTERN TIME	RECORDED RESULT
November 9 · 2:00 a.m.	Successful
November 10 · 2:00 a.m.	Capacity guard refused
November 11 · 2:00 a.m.	Capacity guard refused

The prior successful job passed with an estimate of 50.2 GiB. The inspected logs establish the increase in estimated requirements. They do not identify which individual files caused it.

Next operational action

Resolve the destination capacity shortfall, verify a subsequent scheduled guest backup, and record an isolated restore test.

Separate capacity checks

The media filesystem and local backup destination have distinct limits. Free media space does not establish backup destination capacity.

Backup coverage

Guest archives and media stack configuration snapshots have separate coverage and timestamps. The dashboard exposes guest job and archive status.

AVAILABILITY SNAPSHOT & EVIDENCE

Operational checks and remaining work.

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Docker containers running

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Current HTTP checks passed

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Integrations passed during dashboard work

WORKSTREAM	RECORDED VERIFICATION
Private access	Tailscale connectivity and Jellyfin HTTP response after repair
Dashboard	Desktop/mobile rendering, live widgets, and a working Jellyfin destination
Legacy retirement	Docker mount review, service continuity, and system filesystem recovery
Media cleanup	Exact record matching, file deletion, other-record preservation, and catalog refresh
Current availability	HTTP 200 from Homarr, Jellyfin, Seerr, Radarr, Sonarr, Prowlarr, qBittorrent, Bazarr, and Cleanuparr
Backup status	Inspected task history confirms capacity refusals on November 10 and 11

Follow-up work

1. Restore a successful scheduled backup by resolving destination capacity.
2. Perform an isolated restore test and record the result.
3. Review the retained pre-migration snapshot's recovery value.
4. Continue publishing verified changes and incident outcomes to GitHub.

HTTP checks establish response availability. Off-site playback, transcoding performance, and full backup restoration remain separate validation tasks. Storage figures are point-in-time observations.

Evidence basis · Recorded maintenance results and fresh container, service, filesystem, and Proxmox task-history checks. Publication uses aggregate results and sanitized architecture details.

Primary homelab documentation ↗
Maintenance change log ↗